

Curriculum Vitae

Personal

Name: Aleksander Sandro GRM
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Education and Academia

1. Diploma in Maritime Engineering - Navigation – 1994
Faculty for Maritime Technology and Transportation
University of Ljubljana, Slovenia
Graduate thesis: *Calculation of hydrodynamic parameters for cargo ship long 120m*,
supervisor: MSc. Armand Klančič
2. Diploma in Physics – 2001
Faculty for Mathematics and Physics, Department of Physics
University of Ljubljana, Slovenia
Graduate thesis: *Simulation of ship resistance in viscous media*,
supervisor: Prof. Dr. Rudolf Podgornik
3. MSc in Applied Mathematics (Modelling and Simulation of Complex Realities) – 2003
Joint MSc program between ICTP & SISSA, Trieste, Italy
Completed 1st year (all master courses and 4 months research project)
Project title: *Modelling compressor valves in household refrigerator*,
supervisor: Prof. Dr. Thomas Götz
4. PhD in Industrial Mathematics – 2007
RPTU Kaiserslautern-Landau, Department of Mathematics, Kaiserslautern, Germany
title: *Mathematical Analysis of Macroscopic Models for Slow Dense Granular Flow*
Supervisor: Prof. Dr. Axel Klar and Prof. Emer. Dr. Helmut Neunzert
PhD scholarship: Fraunhofer ITWM
5. Official elections
University of Ljubljana, Faculty of Maritime Technology and Transport
→ *Assitant professor in maritime technology*
University of Ljubljana, Faculty of Mathematics and Physics
→ *Assitant professor in applied mechanics*

Research Interest

- Marine hydromechanics (Theory & Numerics)
- Industrial mathematics (Numerics for PDE's)

Language proficiency

Language	CEFR
English	C1
Italian	B2
German	A2

Work experience

1. 1991-1992: Splošna plovba Piran, Portorož, Slovenia
Deck cadet (junior officer), 15 months of navigation on a Ocean going ship
Obtained Certificate for Officer in charge of navigational watch - OOW (1993, renovated in 2017 with no limits)
2. 2000-2001: HarphaSea Ltd., Koper, Slovenia
programmer on projects:
 - hydrographic applications (C++, autocad)
 - tracking vehicle equipped with GNSS & VHF Radio (C++)
 - geodesy with GPS (C++)
3. 2001-2002: Marine Biology Station, Piran, Slovenia
part time researcher on project *Oceanographic buoy*:
creation of data models and data base (MySQL) from acquired oceanographic data, data analysis and visualisation, data modelling (C++, MatLab®)
4. 2002-2003: University of Ljubljana, Faculty of Maritime Studies and Transportation (FPP), Slovenia
- Assistant lecturer (tutorials in mathematics and mechanics for maritime engineers)
5. 2007-2008: University of Nova Gorica, Slovenia
- Research assistant (tutorials in linear algebra and analysis for physicists and engineers)
6. 2008-2015 : Centre for Computational Continuum Mechanics (C3M), Slovenia
 - programmer: development of variety of codes in scientific computing, data visualisation, web services. Development performed in: C++, Java, C#, Python®, MatLab®, Wolfram Mathematica®
 - project leader, leading R&D (CSO)
 - involved in projects:
 - micro nano satellites propulsion analysis (cold and hot gas) (modelling of compressible fluid flow with slip boundary conditions)
 - extrusion of aluminium and recrystallization (modelling non-Newtonian fluid flow and recrystallization at the extrusion exit point)
 - electro-poration of tumour lesion (development of electro poration model and optimization strategies for optimal tumour treatment)
 - dissolution of stones in Karsic caves (incompressible fluid flow with particulate transport on a moving mesh)
 - assessment of critical parameters in the procedure of cutting stone blocks (numerical modelling of fluids, solids and their interaction)
 - inclusion of aluminium scrap in manufacturing of aluminium alloy (stochastic linear programming of mechanical properties for the scarp added aluminium alloy)
7. 2013 - 2017: University of Ljubljana, Faculty of Maritime Studies and Transportation (FPP), Slovenia
- Research assistant (engineering mechanics and mathematics)
8. 2018 - : University of Ljubljana, Faculty of Maritime Technology and Transportation (FPP), Slovenia
- Assistant professor (engineering mechanics and mathematics, marine hydrodynamics)
9. 2021 - : University of Ljubljana, Faculty of Engineering (FS), Slovenia
- Researcher (computational mechanics)

Computer knowledge

- *Programming*: C++, Java, C#, Python®, MatLab®, Octave®, Wolfram Mathematica®
- *IDE*: Qt Creator, Eclipse, Visual Studio (Subversion, Git, Track)
- *Database*: MySQL
- *OS*: Linux (advanced), Mac, Windows
- *Software*: familiar with variety of scientific computing software/libraries (BLAS, LAPACK, IMKL, GSL, MPI, OpenMP, Deal.II, Fenics/Dolfin, FreeFEM++, OpenFOAM, VTK, ITK, NumPy, SciPy, SymPy, Matplotlib, ...), Ansys-Fluent, Ansys-Mechanical, Ansys-CFX, LS-Dyna; 3D modelling: FreeCAD, SolidWorks, Rhino
- *Server/Client applications*: development of WebServices in Java and Python® for the needs of distributed scientific computing support

Bibliography

- [Scopus - link](#)
- [Google Scholar - link](#)
- Conference contributions
 1. Foundations of computational mathematics, Santander, 2005 (poster session)
 2. Recent advances in non-linear partial differential equations and applications, Toledo, 2006 (poster session)
 3. International conference on mathematical modelling and applications to industrial problems, Calicut, India, 2011 (invited speaker)
 4. Modelling Week and Study Group Meeting on Industrial Problems, Calicut, India, 2013 (resource person)
 5. International Forum for Mathematical Modelling, Colombo, Sri Lanka, 2014 (invited lecturer, resource person)
 6. MARINSTRUCT 2019, *Structural impact of circular buckling fenders on tanker hull* (presentation + paper)
 7. 8th Conference on Marine Technology 2019, Invited talk, *Analysis of generic IGBEM for lifting hydrofoils* (paper)
 8. ASHPC 2026, *Open-source framework for parametric study of hydrofoil profiles and motivation for using Physics-Informed Neural Networks (PINN)* (poster)

Academic visits

1. Department of Marine Technology, NTNU, Trondheim, Norway, 2019/03 - 4 months
2. Faculty of Engineering and Naval Architecture, UNIZG, Zagreb, 2022/04 - 1 week
3. Faculty of Mechanical, Maritime and Materials Engineering, TU Delft, Netherlands, 2022/11, 2 weeks

Projects involved

1. Nanostructured Magneto-electric and Multiferroic Systems, 2008-2011, Slovenian Research Agency
2. M5 modelling in micro/nano satellites, 2009-2012, Slovenian Research Agency
3. Development and validation of a software for numerical modelling of in vivo electroporation - in silico electroporation, 2011-2014, Slovenian Research Agency & FP7
4. Development and application of advanced numerical and experimental methods in study of karst processes, 2011-2014, Slovenian Research Agency & FP7
5. Radial forging process, 2013, GFM, Austria (industrial)
6. Development of aluminium extrusion platform, application to non-Newtonian fluids and re-crystallization, 2012, Impol Ltd., Slovenia (industrial)
7. Stochastic modelling in aluminium alloy production using aluminium scrap: optimal aluminium alloy production and aluminium alloy mechanical properties, 2014, Impol Ltd, Slovenia (industrial)
8. Navigability study of petrol terminal in port of Koper: Material analysis of flexible dolphin structures, Port of Koper, 2016 (industrial)
9. Joint project with yacht designer Andrej Justin on Maxi Sailing Boat - Justin 100, CFD and FEM computations, 2018 (industrial)
10. Joint project with yacht designer Andrej Justin on Catamaran Power Boat - KingCat 51, Power predictions and structural computations (CFD & FEM), 2018 (industrial)
11. Joint project with yacht designer Andrej Justin on Catamaran Power Boat - KingCat 51, Power predictions and structural computations for the foil assisted vessel (CFD & FEM), 2018 (industrial)
12. Lifting foil optimal shape design in the performance of foil assisted vessels, 2019 (academic)
13. PRACE – Dense Linear Algebra peta-scale parallelization, 2019 (academic)
14. SMARTSEA – Erasmus+ K2, Surveillance with underwater vehicles, Pilot master study, Project Leader @ ULFPP, 2019-2022 (academic)
15. Blue4Seas – Erasmus+ K2, Oil spill surveillance, Pilot master study, 2020-2022 (academic)
16. Redesign of Calafuria power boat, Stability and CFD analysis for new power boat design, 2025 (industrial)
17. Power prediction analysis of AUV, Resistance and propeller design with mixed approach using OpenPROP and CFD for special niche AUV in surveillance missions, 2025 (industrial)
18. Yacht and Small Craft Surveyor, Technical survey of Yachts and Small Crafts, 2022– (industrial)